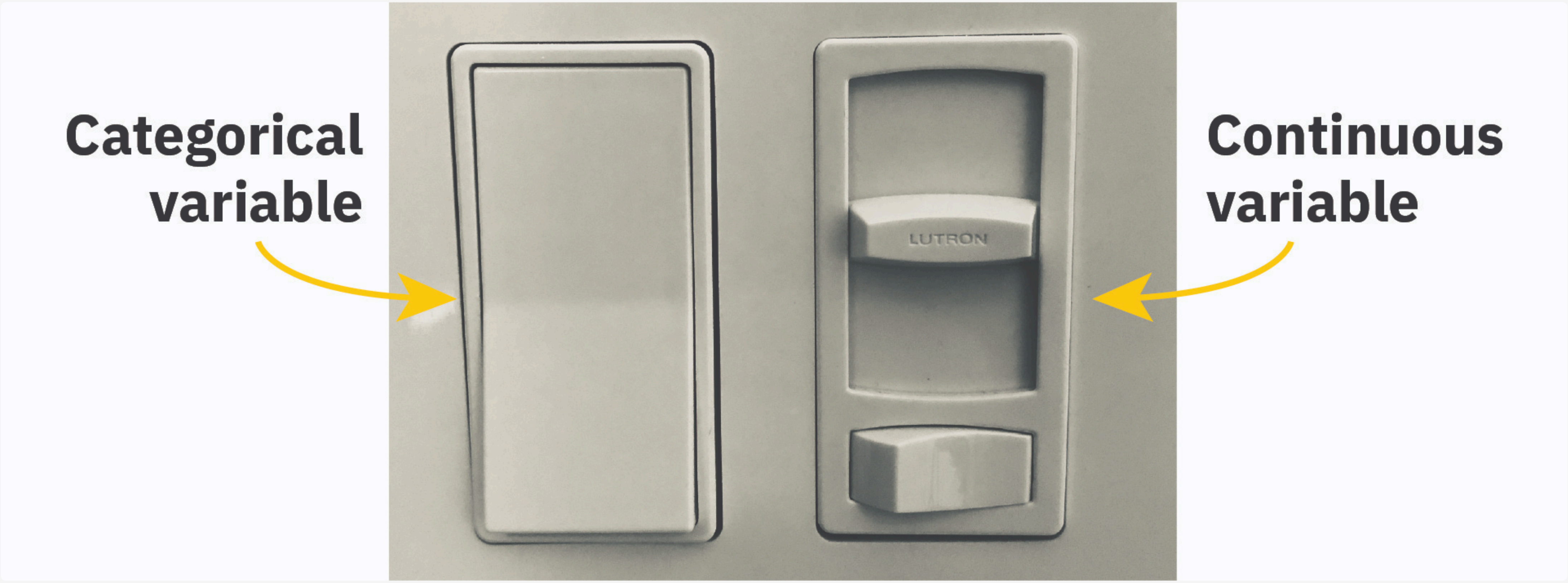
Causal inference with conjoints

Sliders and switches



Categorical variable
LUTRON
Continuous variable

Many simultaneous continuous variables

Many simultaneous categorical variables

```r
library(tidyverse)
library(marginaleffects)
library(parameters)

penguins <- penguins |> drop_na(sex)

model1 <- lm(body_mass ~ flipper_len, data = penguins)
model2 <- lm(body_mass ~ flipper_len + species, data = penguins)
```

```r
model_parameters(model1)
```

```
Parameter   | Coefficient |     SE |                95% CI | t(331) |      p
--------------------------------------------------------------------------
(Intercept) |    -5872.09 | 310.29 | [-6482.47, -5261.71] | -18.92 | < .001
flipper len |       50.15 |   1.54 | [   47.12,    53.18] |  32.56 | < .001
```

```r
model_parameters(model2)
```

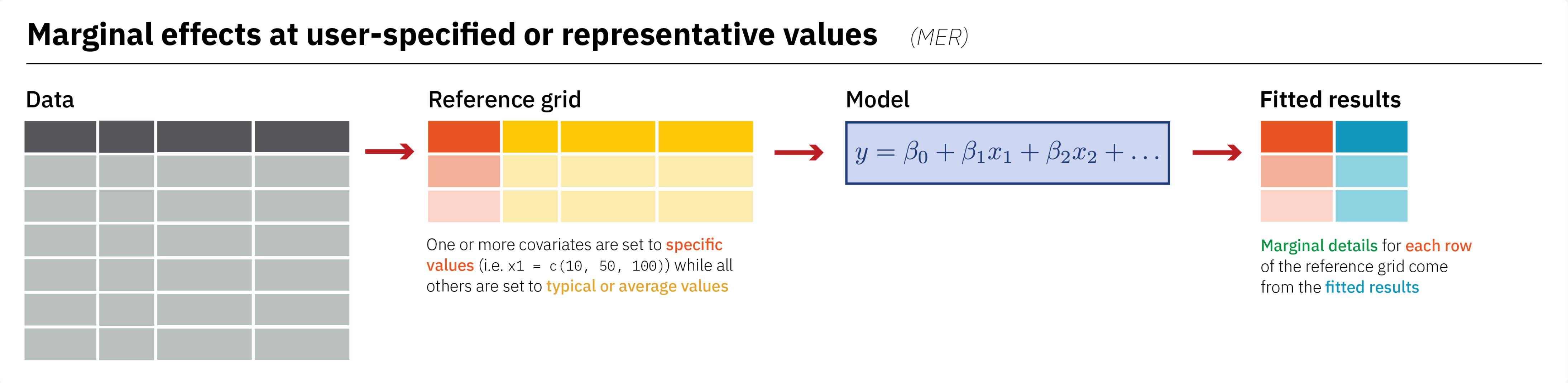
```
Parameter          | Coefficient |     SE |                95% CI | t(329) |      p
---------------------------------------------------------------------------------
(Intercept)        |    -4013.18 | 586.25 | [-5166.44, -2859.92] |  -6.85 | < .001
flipper len        |       40.61 |   3.08 | [   34.55,    46.66] |  13.19 | < .001
species [Chinstrap] |     -205.38 |  57.57 | [ -318.62,   -92.13] |  -3.57 | < .001
species [Gentoo]   |      284.52 |  95.43 | [   96.79,   472.25] |   2.98 | 0.003
```

{marginaleffects}

> Quit working with raw coefficients!

Arel-Bundock, Vincent, Noah Greifer, and Andrew Heiss. 2024. "How to Interpret Statistical Models Using marginaleffects for R and Python." Journal of Statistical Software 111 (9): 1–32. https://doi.org/10.18637/jss.v111.i09.

Arel-Bundock, Vincent. 2026. Model to Meaning: How to Interpret Statistical Models with R and Python. A Chapman & Hall Book. CRC Press, Taylor & Francis Group. https://doi.org/10.1201/9781003560333, https://marginaleffects.com/



1. **Quantity**: What is the quantity of interest? Do we want to report a prediction or a function of predictions (average, difference, ratio, derivative, etc.)?
2. **Grid**: What predictor values are we interested in? Do we want to report estimates for the units in our dataset, or for hypothetical or representative individuals?
3. **Aggregation**: Do we report estimates for every observation in the grid or a global summary?
4. **Uncertainty**: How do we quantify uncertainty about our estimates?
5. **Test**: Which (non-)linear hypothesis or equivalence tests do we conduct?

```r
model_parameters(model1)
```

```
Parameter   | Coefficient |     SE |                 95% CI | t(331) |      p
----------------------------------------------------------------------------
(Intercept) |    -5872.09 | 310.29 | [-6482.47, -5261.71] | -18.92 | < .001
flipper len |       50.15 |   1.54 | [   47.12,    53.18] |  32.56 | < .001
```

```r
avg_comparisons(model1)
```

```
 Estimate Std. Error    z Pr(>|z|)     S 2.5 % 97.5 %
     50.2       1.54 32.6   <0.001 770.2  47.1   53.2

Term: flipper_len
Type: response
Comparison: +1
```

```r
model_parameters(model2)
```

```
Parameter           | Coefficient |     SE |                95% CI | t(329) |      p
------------------------------------------------------------------------------------
(Intercept)         |    -4013.18 | 586.25 | [-5166.44, -2859.92] |  -6.85 | < .001
flipper len         |       40.61 |   3.08 | [   34.55,    46.66] |  13.19 | < .001
species [Chinstrap] |     -205.38 |  57.57 | [ -318.62,   -92.13] |  -3.57 | < .001
species [Gentoo]    |      284.52 |  95.43 | [   96.79,   472.25] |   2.98 | 0.003
```

```r
avg_comparisons(model2)
```

```

        Term           Contrast Estimate Std. Error     z Pr(>|z|)     S  2.5 % 97.5 %
 flipper_len +1                     40.6       3.08 13.19  < 0.001 129.5   34.6   46.6
 species     Chinstrap - Adelie   -205.4      57.57 -3.57  < 0.001  11.4 -318.2  -92.5
 species     Gentoo - Adelie       284.5      95.43  2.98  0.00287   8.4   97.5  471.6

Type: response
```

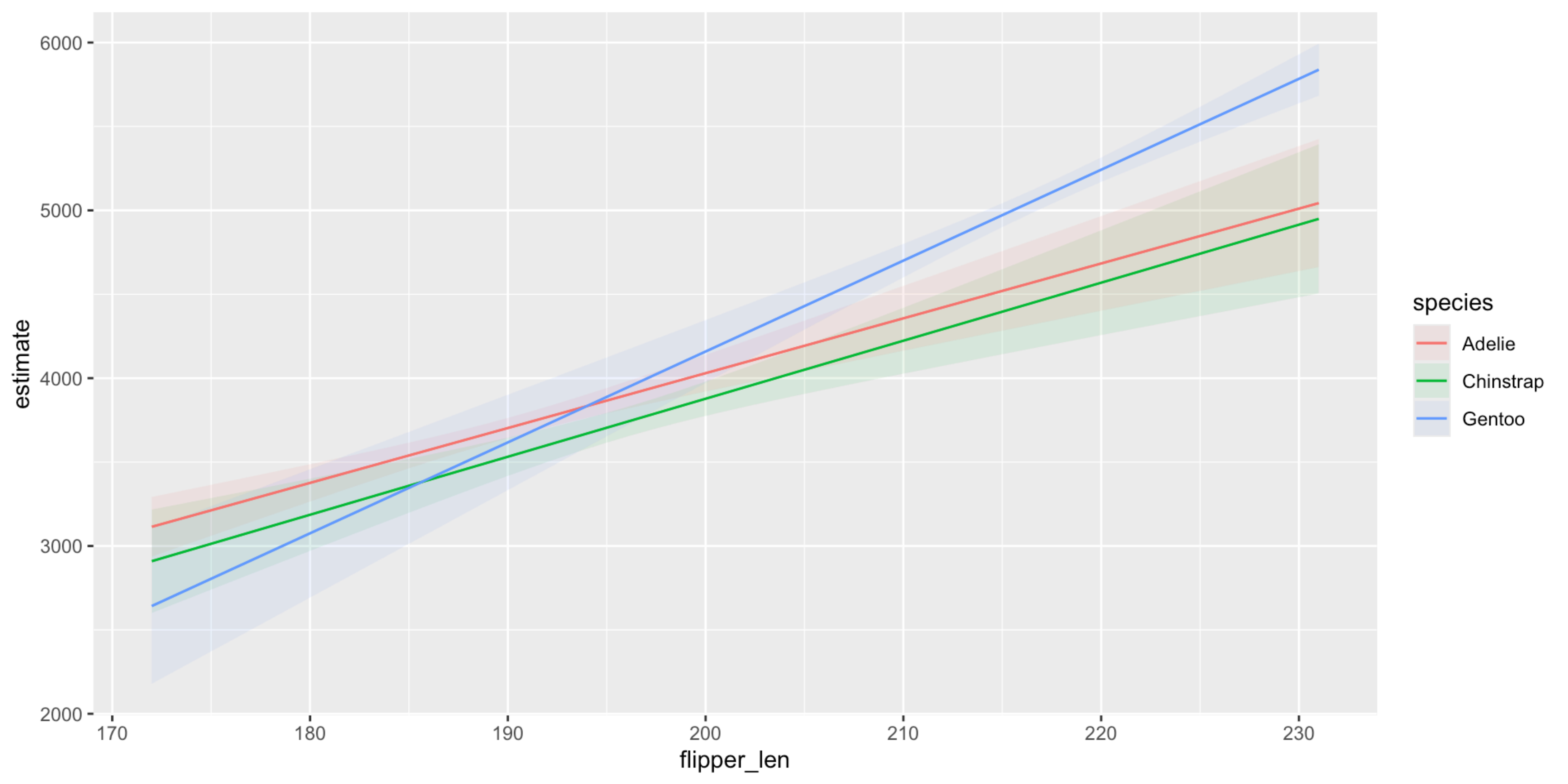
```r
model3 <-lm(body_mass ~ flipper_len * species, data = penguins)
model_parameters(model3)
```

```
Parameter                         | Coefficient |      SE |                 95% CI | t(327) |
p
-------------------------------------------------------------------------------------------------
--
(Intercept)                       |    -2508.09 |  892.41 | [-4263.68,  -752.49] |  -2.81 |
0.005
flipper len                       |       32.69 |    4.69 | [   23.46,    41.92] |   6.97 | <
.001
species [Chinstrap]               |     -529.11 | 1525.11 | [-3529.37,  2471.16] |  -0.35 |
0.729
species [Gentoo]                  |    -4166.12 | 1431.58 | [-6982.39, -1349.84] |  -2.91 |
0.004
```

```r
avg_comparisons(model3, by = "species")
```

```
        Term           Contrast   species Estimate Std. Error      z Pr(>|z|)    S  2.5 % 97.5 %
 flipper_len +1                   Adelie      32.7       4.69  6.967  < 0.001 38.2   23.5   41.9
 flipper_len +1                   Chinstrap   34.6       6.31  5.478  < 0.001 24.5   22.2   46.9
 flipper_len +1                   Gentoo      54.2       5.15 10.516  < 0.001 83.5   44.1   64.3
 species     Chinstrap - Adelie Adelie      -170.9      65.04 -2.627  0.00861  6.9 -298.3  -43.4
 species     Gentoo - Adelie    Adelie       -83.4     146.97 -0.567  0.57052  0.8 -371.4  204.7
 species     Chinstrap - Adelie Chinstrap   -160.1      60.39 -2.651  0.00803  7.0 -278.4  -41.7
 species     Gentoo - Adelie    Chinstrap     39.5     122.28  0.323  0.74675  0.4 -200.2  279.2
 species     Chinstrap - Adelie Gentoo      -119.7     193.37 -0.619  0.53580  0.9 -498.7  259.3
 species     Gentoo - Adelie    Gentoo       499.3     135.18  3.694  < 0.001 12.1  234.4  764.3
```

```
plot_predictions(model3, condition = c("flipper_len", "species"))
```



What are marginal means?

OLS with categorical predictors

```r
model <- lm(body_mass ~ species + sex, data = penguins)
```

Predictions across balanced grid

```r
grid <- datagrid(model = model, species = unique, sex = unique)

preds <- predictions(model, grid)
```

```r
preds |>
  as_tibble() |>
  select(species, sex, estimate)
```

```
# A tibble: 6 × 3
  species   sex    estimate
  <fct>     <fct>     <dbl>
1 Adelie    female    3372.
2 Adelie    male      4040.
3 Chinstrap female    3399.
4 Chinstrap male      4067.
5 Gentoo    female    4750.
6 Gentoo    male      5418.
```

	Sex	
Species	**Female**	**Male**
Adelie	3372	4040
Chinstrap	3399	4067
Gentoo	4750	5418

Means in the margins

	Sex		
Species	**Female**	**Male**	**Marginal mean**
Adelie	3372	4040	3706 (3372 + 4040) / 2
Chinstrap	3399	4067	3733 (3399 + 4067) / 2
Gentoo	4750	5418	5084 (4750 + 5418) / 2
Marginal mean	3841 (3372 + 3399 + 4750) / 3	4508 (4040 + 4067 + 5418) / 3	

> For the sex-specific averages, all the between-species variation is “marginalized out” or accounted for; for the species-specific averages, all the between-sex variation is similarly marginalized out.

Differences in marginal means

	Sex			
Species	**Female**	**Male**	**Marginal mean**	**Δ**
Adelie	3372	4040	3706 (3372 + 4040) / 2	—
Chinstrap	3399	4067	3733 (3399 + 4067) / 2	27 Chinstrap−Adelie
Gentoo	4750	5418	5084 (4750 + 5418) / 2	1378 Gentoo−Adelie
Marginal mean	3841 (3372 + 3399 + 4750) / 3	4508 (4040 + 4067 + 5418) / 3		
Δ	—	668 Male−Female		

Coefficients and predictions

```r
parameters::model_parameters(model)
```

```
Parameter           | Coefficient |    SE |              95% CI | t(329) |      p
---------------------------------------------------------------------------------
(Intercept)         |     3372.39 | 31.43 | [3310.56, 3434.21] | 107.31 | < .001
species [Chinstrap] |       26.92 | 46.48 | [ -64.52,  118.37] |   0.58 | 0.563
species [Gentoo]    |     1377.86 | 39.10 | [1300.93, 1454.78] |  35.24 | < .001
sex [male]          |      667.56 | 34.70 | [ 599.29,  735.82] |  19.24 | < .001
```

```r
avg_predictions(model, by = "species", newdata = "balanced")
```

```

   species Estimate Std. Error     z Pr(>|z|)   S 2.5 % 97.5 %
 Adelie        3706       26.2 141.4   <0.001 Inf  3655   3758
 Chinstrap     3733       38.4  97.2   <0.001 Inf  3658   3808
 Gentoo        5084       29.0 175.1   <0.001 Inf  5027   5141

Type:  response
```

```r
avg_predictions(model, by = "sex", newdata = "balanced")
```

```

    sex Estimate Std. Error   z Pr(>|z|)   S 2.5 % 97.5 %
 female     3841       25.3 152   <0.001 Inf  3791   3890
 male       4508       25.1 180   <0.001 Inf  4459   4557

Type:  response
```